

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Digital Forensics and Information Security

Semester - II - April-2026

Subject Code: CTMSDFIS24 SII P1

Date: 24/04/2026

Subject Name: Network Security and Forensics

Time: 2.30 PM TO 5.30 PM

Total Marks: 100

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

		Marks
Q.1	Attempt any three.	
(a)	Discuss the working of Transmission Control Protocol (TCP). Explain how TCP ensures reliability using acknowledgments, sequencing, and retransmission.	08
(b)	Define a bridge and explain its functionality in segmenting a network. How does it use MAC addresses to filter traffic?	08
(c)	Discuss the importance of DHCP in modern networks. Describe the DORA process also.	08
(d)	Explain IP address classes and determine how many devices can be connected in a given university if Class C IP address assigned to it?	08
Q.2	Attempt any three.	
(a)	Explain the complete Penetration Testing Life Cycle in detail. Discuss each phase including Scope definition, Statement of Work (SOW), Reconnaissance, Target Enumeration, Vulnerability Identification, Assessment, Exploitation, and Reporting.	08
(b)	What is SNMP (Simple Network Management Protocol)? Explain how SNMP services can be exploited during penetration testing.	08
(c)	Explain the concept of Scanning in penetration testing. Also, describe the working of ICMP (Ping) scanning and its significance in network discovery.	08
(d)	Explain the concept of network scanning in penetration testing. Also explain different port states (open, closed, filtered, unfiltered, open filtered, closed filtered) and their significance in identifying network vulnerabilities.	08

Q.3

Attempt any three.

- (a) What are Digital Signatures? Explain their working process and importance in ensuring authentication, integrity, and non-repudiation in electronic communication. **08**
- (b) Explain the concept of Hash Functions and Message Digest. How do they differ from encryption? **08**
- (c) Explain the concept of encryption in cryptography. Differentiate between symmetric and asymmetric encryption techniques with suitable examples and diagrams. **08**
- (d) Explain Caesar Cipher algorithm for encryption and decryption process. Encrypt the plaintext "WAR BETWEEN USA AND IRAN" using a Rail Fence Cipher with 3 rails. **08**

W A R
B E T W E E N
U S A A N D I R A N

Q.4

Attempt any two.

- (a) What is a Wireless Access Point (WAP)? Explain its role in WLAN architecture. **07**
- (b) Describe the authentication and encryption mechanisms in WLANs. **07**
- (c) Explain how wireless packet analysis can be used in forensic investigations **07**

Q.5

Attempt any two.

- (a) What is Network-Based Digital Evidence? Explain its significance in investigating cyber incidents. **07**
- (b) Explain the process of Evidence Acquisition in network forensics. What are the key challenges in collecting volatile and non-volatile network data while maintaining integrity? **07**
- (c) Discuss the importance of key management in symmetric and asymmetric encryption. **07**